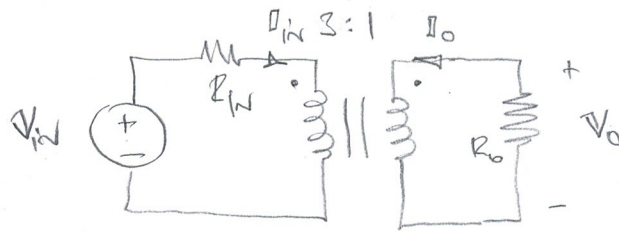


7.46)



$$V_{in} = 240 \angle 0^\circ$$

$$R_{in} = 50 \Omega$$

$$R_o = 20 \Omega$$

a) Bestäm strömmen på primärsidan

Lösning: Transformera impedans

$$Z_1 = \frac{1}{n^2} Z_2$$

$$Z_1 = 3^2 R_o = 9 \cdot 20 = \underline{180 \Omega}$$

Strömmen på primärsidan blir med KVL

$$-V_{in} + (R_{in} + Z_1) I_{in} = 0 \Rightarrow I_{in} = \frac{V_{in}}{R_{in} + Z_1} = \frac{240 \angle 0^\circ}{50 + 180} = \underline{1.043 \angle 0^\circ A}$$

b) Spänning över resistansen på sekundärsidan

Lösning: Vi vet att

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \Rightarrow V_2 = \frac{N_2}{N_1} V_1$$

DVS

$$V_2 = \frac{1}{3} V_1 = \frac{1}{3} (V_{in} - R_{in} I_{in}) =$$

$$= \frac{1}{3} (240 \angle 0^\circ - 50 \cdot 1.043 \angle 0^\circ) =$$

$$= \underline{62.62 \angle 0^\circ \text{ V}}$$

c) Effekten som levereras på sekundärsidan

Lösning: Effekt

$$P = \frac{1}{2} V I^*$$

DVS

$$P = \frac{1}{2} V_2 I_2^* = \frac{1}{2} V_2 \frac{V_2^*}{Z_2^*} = \frac{1}{2} |V_2|^2 \frac{1}{Z_2^*} =$$

$$= \frac{1}{2} 62.62^2 \frac{1}{20} = \underline{98.02 \angle 0^\circ \text{ VA}}$$

eller som

$$\underline{P = 98.02 \text{ WATT}}$$

d) VERKLEINERUNGSGRAD?

Lösung:

$$\frac{P_{IN}}{P_0} = \frac{|\Pi_N|^2 R_{IN}}{|\Pi_0|^2 R_{UT}} = \frac{1.043^2 50}{(3 \cdot 1.043)^2 20} \approx \underline{\underline{0.278}}$$

SVAR: a) $\Pi_N = 1.04 \angle 0^\circ \text{ A}$

b) $V_2 = 62.6 \angle 0^\circ \text{ V}$

c) $P = 18.0 \text{ W}$

d) $\frac{P_{IN}}{P_0} = 0.28$